Project decision and performance

Measured simulation performance • implementation audit • staged upgrade plan

Decision

The current project is a working simulation prototype with coordinated morphology and drive control. The low-cost visible preset completed 208.5 simulated seconds at mean RTF 0.998, with sampled one-second RTF p05 of 0.994 and a 151 ms forward response trial. This demonstrates near-real-time operation for that tested configuration. A repeatable hard real-time guarantee and hardware-calibrated predictive accuracy remain unestablished. Use this report as the implementation plan for the next revision.

What changed in this review

The quality-oriented default camera is now 480 x 270 at 10 Hz. Optional image degradation processing is disabled by default and can be enabled for sensor experiments. Arena visuals and the 2 ms physics timestep are retained. A workspace launch lock prevents accidental duplicate simulations. An optional run_realtime.sh preset lowers the sensor camera to 320 x 180 at 5 Hz and disables scene shadows, and defaults LP_NUM_THREADS to 2 for Mesa software rendering, while retaining the same dynamics. A longer benchmark records simulation time, wall time, response, combined movement, temperature and trajectory samples.

Configuration	Sim time (s)	Mean RTF	1 s RTF p05*	Response (ms)
Visible / previous camera	58.5	0.947	0.853	140
Visible / balanced	58.5	0.960	0.923	149
Visible / longer soak	208.5	0.927	0.899	153
Visible / low-cost preset	208.5	0.932	0.903	141
Visible / 2 render workers	208.5	0.998	0.994	151
Headless / 1 ms	58.5	0.975	0.937	130
Headless / 2 ms	58.5	0.999	0.997	131

*Sampled/interpolated application-level estimate; one response trial per case. Source: local validation/performance JSON traces, 28 October 2026.

Sustained-performance finding

The balanced visible run averaged 0.960 RTF over 58.5 simulated seconds, but a 208.5-second run averaged 0.927 with a sampled one-second fifth percentile of 0.899. That balanced run misses the proposed thresholds. Disabling shadows and reducing the camera alone averaged only 0.932. The otherwise matching two-render-worker preset averaged 0.998 over 208.5 simulated seconds and passed the proposed mean/p05 performance thresholds in this run. Repeated runs, latency p95, ten-minute loaded testing and console/display-rate checks remain open. These sequential observations do not isolate all system-load effects.

Implementation inventory

Implemented and exercised

ROS 2 controllers, chassis expansion/compaction, ride height, four passive suspension joints, four-wheel swerve kinematics, combined drive/strafe/yaw plus expansion, current-limited drive effort, battery and motor thermal model, quantized encoder output, IMU output, native contact sensors, a combined keyboard console and thirteen connected sector entrances.

Implemented with simplifications

The 10 kg mass and component inertias are assumed. The 150 mm diameter wheels are a proposed mechanical revision from the earlier 80 mm wheels. Tyre contact uses a smooth rigid cylinder. Steering, height and morphology use position servos. Coil visuals remain rigid while barrel and piston move on separate links. The upper mechanism uses an open chain with controlled visual closure.

Diagnostic predictions

Pacejka forces, Bekker sinkage/drag, patch pressure, loaded tyre radius, per-wheel normal loads, step-loss risk and tip margins are reported as predictions or estimates. They are not all applied to Gazebo dynamics. The drive motor model and native suspension forces are applied; detailed steering electronics and deformable soil are not.

Environment status

Nine terrain columns cover smooth road, grass, sand, mud, ramps, debris, rocks, snow and mixed terrain. Connected village, industrial/pipeline, rubble and depot sectors are present. Source meshes and textures are local; the surrounding elevation is an engineered derivative of a survey DEM. This is a constructed test facility, not a geometrically exact reconstruction of a real site.

Verification status

The latest full software suite reports 51 checks with zero failures. Historical runtime checks passed all thirteen safe entrances with the revised wheel geometry and drive effort mode. These entrance checks do not verify each full terrain lane. Moving emergency-stop stopping distance, complete slope/step traversal and physical hardware comparison still need dedicated tests.

Physics and mechanical findings

P0: freeze the proposed hardware baseline

Confirm wheel diameter/width, loaded radius, wheel mass, motor/gearbox identity, battery and steering arrangement before tuning. The original specification used a 63 mm radius placeholder; the current model uses 75 mm. Changing launch wheel radius alone does not update the diagnostic parameter file, so non-default geometry can invalidate encoder, torque and contact predictions.

P0: one source of physical parameters

Several suspension values exist both in params.yaml and the URDF/native plugin. Seal friction, bump stiffness and force caps are also hard-coded in the plugin. Generate URDF/plugin settings and diagnostic settings from one validated schema, record units and uncertainty, and reject conflicting launch overrides. Export a resolved parameter manifest with each run.

P0: reconcile surface physics and diagnostics

The rocks sector is labeled wet_rock in the diagnostic model, where static friction is 0.35, while its ground SDF uses approximately 0.70/0.665. Rock obstacle meshes have their own material friction. Generate all surface coefficients from one table and verify the effective engine behavior using slip/pull tests. A visual material is not a friction calibration.

P1: tyre/soil model selection

Add measured rolling resistance first, then a validated tyre model with regularized combined slip and relaxation. A new tangential force model must replace or deliberately coordinate native friction to avoid double counting. Add a calibrated soft-soil drag approximation only for the surfaces being studied. Deformable tyres, ruts and snow compaction are separate, substantially larger developments.

P1: suspension and loads

Measure spring force versus travel, ring-down response, damping versus velocity and temperature, seal friction and bump stops. Current damping is a lumped passive model with an assumed heat capacity/cooling law. Use contact-wrench sensing or validated force reconstruction for normal load; spring displacement alone is an estimate. Chassis flex and bush compliance remain absent.

Design feasibility bound

The current drive limit is 0.5 N m per wheel. With a 75 mm wheel radius and 10 kg total mass, ideal total drive force is 26.67 N and the zero-rolling-resistance grade bound is about 15.8 degrees. A 25-degree steady climb needs about 0.80 N m per wheel with Crr = 0.015, before extra drivetrain losses. Larger ramps are useful failure tests; they cannot all be promised climbable with this motor limit.

Controls, sensors and reliability

P0: prove the stop and watchdog path

The existing benchmark latches emergency stop after the robot has already braked. Add a moving emergency-stop test, focus-loss and key-release tests, stale-command/state tests and a control-node crash test. Measure stopping time and distance on high- and low-friction ground. Require fresh measured joint states before any actuator command.

P0: inspect controller limit messages

The runtime logs still contain throttled command-limit messages during stage movement and steering transitions. Trace the requested and clamped position/velocity commands, check controller timing and trajectory lead, and verify tracking error under combined motion. Passing the functional test does not resolve these messages. Keep actuator limits intact while correcting the command trajectory.

P1: actuator realism

Drive torque uses back EMF, current limiting, gear efficiency, battery sag and winding/driver heating. Steering, ride height and expansion remain idealized position-served axes. Add measured steering torque-speed limits, gearbox backlash, stiction, rate limits and step-loss/homing behavior. Include steering/linear-actuator energy in the shared power budget; current battery predictions principally cover drive and an electronics allowance.

P1: encoder-based feedback

Quantized encoder/steering sensor topics are available, but the controller/plant uses raw simulator feedback. Add a configurable sensor-feedback path with timestamped sampling, delay, quantization and dropout. Validate encoder odometry against ground truth on straight, circle and mixed-slip trajectories. Parallel-wheel swerve odometry has observability limitations that should be reported.

P1: timing instrumentation

Record source command stamps, manager receipt, actuator command time, joint response and body response with both monotonic wall time and simulation time. Add ROS callback timing, sensor achieved rates, dropped samples and graphics frame timing. Run at least twenty response trials and five repeated benchmarks before reporting response p95.

P1: sensor realism

Scale camera calibration with resolution and validate CameraInfo against the image stream. Current blur/exposure/dirt/delay are simple image effects; the balanced default disables them. Add camera noise/calibration, IMU bias and vibration transfer, encoder edge-rate limits and measured communication timing only when those effects matter to the chosen experiments.

P2: hardware interface

Create a hardware/simulation interface with matching command units, saturation, timestamps, fault codes and homing state. Then add ESP32/ROS hardware-in-the-loop tests. Network delay, firmware timing and EMI need measured or deliberately injected models; they cannot be inferred from a visually realistic Gazebo scene.

Arena, graphics and presentation

P1: make the facility a benchmark

Add measured side-slope lanes, step/gap ladders scaled to the loaded wheel radius, washboard, potholes/ruts, a split-friction lane, an unpowered coast-down lane and ground markers every metre. Provide safe reset poses and IDs for every fixture. Store fixture dimensions and collision geometry together and verify with automated measurements.

P1: complete the indoor route

The current depot is an open logistics area. The specification also requests a true indoor block with tile/grout, carpet, rubber mat, thresholds, doors, narrow corridors and turns. Add these as a connected section using imported assets and calibrated test fixtures; retain an accessible loop between indoor, semi-outdoor and off-road sectors.

P2: visual quality within a budget

Use higher quality imported assets where the robot camera can see them, correct material scale, more natural blending at terrain boundaries, better building facades, vegetation variation and approved sky/lighting assets. Use distance-based mesh detail and instancing for decoration. Keep photographed/source textures and attribution. Measure RTF after each asset group; visual detail should not consume the control budget.

P2: weather and interaction

Temperature already affects selected physical/diagnostic models. Wetness, lux and scenario labels do not automatically change all native world properties. Add synchronized lighting/day-night and wet-surface changes with a clear event log. Fog/rain/dust can be introduced as perception experiments. Dynamic people and movable obstacles need deterministic paths, reset states and collision tests.

Presentation deliverables

Create a supervised route that shows normal driving, crab motion, in-place yaw, gradual transformation, suspension travel and a bounded obstacle attempt. Overlay measured speed, slip estimate, motor current, damper temperature, body attitude, active sector and RTF. Record a clean walkthrough and a fixed-camera close view of each wheel module. Keep the demo outcome distinct from validation evidence.

Rendering limitation observed

The execution environment logged a Mesa device-query warning, and nvidia-smi could not reach a driver. This is evidence to investigate acceleration access, not proof that the user computer has no working GPU. Verify the actual renderer on the desktop before changing drivers. No driver or operating-system replacement was performed in this review. The lower-cost preset bounds Mesa software rasterization workers through LP_NUM_THREADS=2; hardware rendering drivers may ignore that setting. The observed effect is recorded in a separate benchmark.

Upgrade roadmap and release gates

Stage A — reproducible baseline | first

Consolidate parameters, resolve the rock surface mismatch, validate geometry/parameter agreement, add a launch manifest, preserve source hashes and fix clean shutdown paths. Keep the duplicate-launch guard. Exit: generated URDF, SDF and diagnostic configuration agree; missing/conflicting parameters fail startup; all software and resource checks pass.

Stage B — performance and operational reliability

Add timing instrumentation, achieved sensor-rate monitoring, repeated moving-response/stop tests and a ten-minute loaded soak. Compare visible, headless and high-quality camera settings. Exit target: mean RTF >= 0.95; sampled one-second RTF fifth percentile >= 0.90; response p95 <= 250 ms over at least twenty trials; five repeat runs; no stale-control or shutdown faults. These are proposed acceptance targets, not existing guarantees.

Stage C — calibrated module mechanics

Freeze the wheel/motor/steering design, derive all component mass/inertia consistently, fit drive and steering curves, validate suspension travel, ring-down and thermal behavior. Exit: unit-correct interfaces and parameter uncertainty recorded; measured and simulated curves compared on held-out tests; mechanism closure and interference checked across the full motion range.

Stage D — controlled terrain experiments

Build the indoor and off-road fixture suite, add contact-load/slip instrumentation and calibrated rolling resistance. Evaluate tyre/soil model upgrades only after the rigid-contact baseline is understood. Exit: automated ramp, step/gap, side-slope, drawbar, coast-down and washboard protocols with repeatable reset, declared pass/failure criteria and full data logs.

Stage E — hardware transfer and sensing

Run encoder/IMU/camera calibration, match firmware command/feedback timing, add communication faults and hardware-in-the-loop. Use measured loaded radius, motor efficiency and thermal curves. Exit: the same command trace runs on hardware and simulation; trajectory/current/temperature errors and uncertainty are reported rather than judged by appearance.

Stage F — paper and presentation

Define a focused research question such as morphology-aware swerve control versus a fixed-morphology baseline. Run ablations for attachment-velocity compensation, automatic morphology and actuator limits, with identical routes and seeds. Report energy, slip, pose error, body attitude, success rate and timing. Separate a simulation study with assumptions from a hardware-validated study. Freeze code, assets, parameters and raw data for the submitted results.

Validation plan and research scope

Required motion and fault tests

Straight and reverse runs; circles at several radii; crab and in-place yaw; expansion/compaction while moving; height changes under load; steering at its rate/effort limits; moving emergency stop; key-release/focus-loss; stale command/state; node failure; reset/homing after brownout. Test safe transitions rather than only steady commands.

Required physics tests

Per-surface slip/pull measurements, drawbar pull, ramp/step/gap and side-slope limits, unpowered coast-down, drop/ring-down, washboard speed sweep, motor stall/current-limited response, battery sag, heat-soak/cool-down, and timestep/contact sensitivity. Record failures as results, especially where the 0.5 N m drive limit prevents a climb.

Required sensor tests

Wheel/steering encoder scale and lost counts, odometry drift, IMU stationary bias/noise and vibration spectrum, camera intrinsics and achieved rate, timestamp alignment, exposure/delay under controlled lighting and communication dropout. Ground truth and noisy sensor channels must remain separately labeled.

Numerical convergence

Compare 1 ms and 2 ms integration with matching initial state, command trace, controller rate and seed. Evaluate trajectory, suspension extrema, contact impulses and energy, not only end position. Current short runs are sensitivity observations. They do not establish convergence on obstacles, impacts or soft ground.

Prioritization

Implement Stages A and B before adding autonomous navigation or more decorative assets. Then prioritize measured mechanics and benchmark fixtures. Navigation/SLAM, terrain classification, predictive suspension, traction control and experiment automation are valuable additions after sensor and contact baselines are trustworthy. Add sensors only if they exist on the planned hardware or the experiment explicitly studies them.

Definition of research readiness

A research-ready release has a reproducible protocol, traceable parameters and uncertainty, validated dynamics for its stated operating range, complete raw data, explicit limitations and a defensible baseline comparison. A passing software suite or attractive world is only part of that evidence. Until real measurements exist, label hardware-performance parameters assumed.

How to read the results

RTF = elapsed simulated time / elapsed wall time. Aggregate RTF below is total simulated duration divided by summed wall duration. Response is command publication to ground-truth speed above 0.03 m/s; each case contains one forward-response trial. The sampled one-second RTF fifth percentile is interpolated from application samples at roughly 10 Hz. It is not a count of physics deadline misses.

Limits of the evidence

Runs were sequential rather than randomized, and system load varied. The visible benchmark used automated commands with the control console disabled. It therefore does not measure keyboard-to-motion latency, console overhead, achieved sensor publication rates or display frame rate. A single run per configuration does not establish a latency percentile, long-run reliability or statistical significance. The historical very slow runs were collected under a much higher desktop load and cannot be used as a controlled before/after comparison.

Running, evidence and references

Default presentation configuration

In /home/diya/Documents/Codex/2026-10-24/ac/outputs/morph_spec_v4_ws run: bash run_arena.sh. This uses the combined controls window, drive effort model, 2 ms physics, 480 x 270 camera at 10 Hz and disabled optional camera effects. Keep the control window focused for keyboard input.

Comparison configurations

Reference timestep: bash run_arena.sh physics_step:=0.001. High-quality sensor comparison: bash run_arena.sh camera_rate:=15 camera_width:=640 camera_effects:=true. Lower-cost visible mode: bash run_realtime.sh. Headless experiment: bash run_arena.sh headless:=true controls:=none. Close the current simulation before starting another; the launch lock now rejects duplicates from this workspace.

Reproduce the benchmark

Source /opt/ros/lyrical/setup.bash and install/setup.bash in the same ROS domain and Gazebo partition as the launch. Run python3 src/morph_sim/tools/benchmark_realtime.py with MORPH_BENCH_OUTPUT set to the output JSON path. MORPH_SOAK_SECONDS controls the sustained idle duration. The script sends motion commands; use controls:=none during automated experiments.

Project evidence

Raw benchmark traces: validation/performance/*.json. Aggregated metrics, source hashes, proposed acceptance gates and scope limitations: validation/performance/summary.json. Historical safe-entrance checks: validation/sector_checks.json. New module inertia assumptions: validation/module_inertia_assumptions.json. Prior source backup: backups/before_research_upgrade.tar.gz. These files accompany this report.

Review sources

The supplied Swerve Robot — Interaction & Environment Specification v1.0 (12 pages) was reviewed against the current launch, controller core, URDF/Xacro, native suspension plugin, electrical/thermal runtime, sensor code, world generator and validation records. The older LOCKED_DIAGNOSTIC.md describes an earlier locked-upper diagnostic and is historical context, not the current launch configuration.

Primary technical references

ROS 2 real-time background: https://docs.ros.org/en/rolling/Tutorials/Demos/Real-Time-Programming.html . Gazebo Sensors update-rate API: https://gazebosim.org/api/sensors/9/classgz_1_1sensors_1_1Sensor.html . gz_ros2_control position-interface behavior: https://control.ros.org/jazzy/doc/gz_ros2_control/doc/index.html . Installed software is ROS 2 Lyrical / Gazebo Sim 10; version-specific behavior should be checked against the installed interfaces. Mesa rendering-thread setting: https://docs.mesa3d.org/envvars.html . Asset licenses and provenance remain in ARENA_GUIDE.md and the model SOURCE.json records.